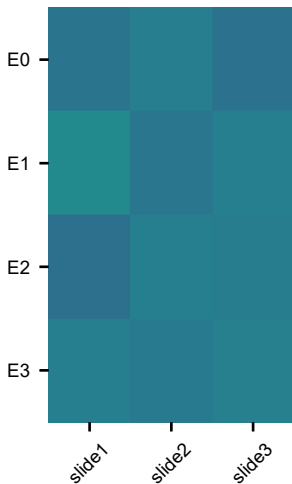
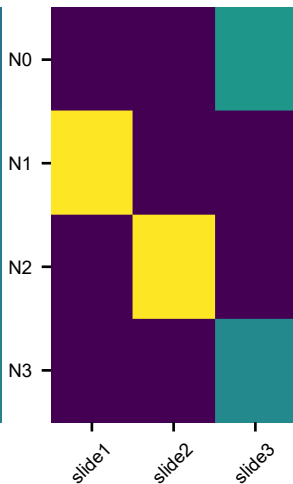


Domain composition per slide (column-wise consistency = transferability)

Expression-only (PCA+KMeans)



Niche smoothing (kNN)



Niche smoothing + per-slide z

